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PRISM: Pruning for Rank-adaptive Interpretable Segmentation Model with Application to Historical Document Multiband Images

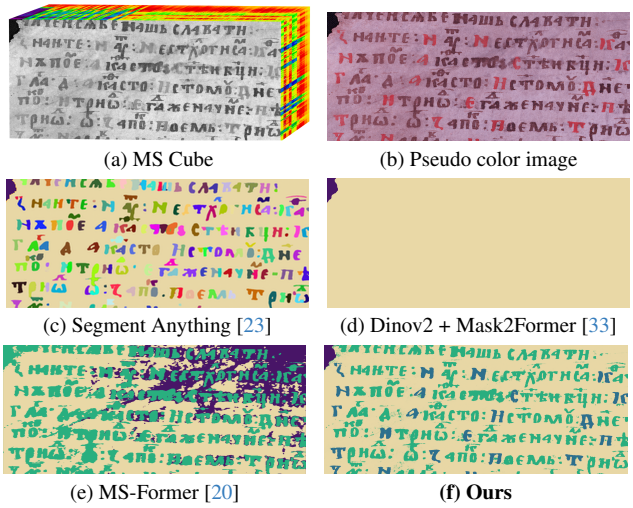
Kilian Declercq[1,2] Abderrahmane Rahiche[1] Mohamed Cheriet[1]
[1]Synchromedia Laboratory, École de Technologie Supérieure (ÉTS), Montréal, Canada
[2]ECAM LaSalle, Lyon, France

Abstract

*Multispectral (MS) imaging reveals latent content in historical documents by leveraging material-specific spectral signatures. Low-rank decompositions such as Nonnegative Matrix Factorization (NMF) effectively extract these components, but selecting the appropriate rank remains an open challenge in unsupervised settings. We propose **PRISM**, a structured autoencoder that embeds a tri-factor NMF into a convolutional architecture and integrates similarity-driven pruning to automatically infer the effective number of components during training. The encoder enforces spatial attention to produce coherent abundance maps, while the decoder reconstructs interpretable spectral signatures through constrained linear decoding. This formulation yields compact, interpretable representations with no supervision or manual rank tuning. A Minimum Description Length criterion estimates the optimal rank directly from the input cube, balancing model complexity with data reconstruction. Experiments on historical manuscripts and cross-domain remote sensing datasets demonstrate PRISM's ability to produce meaningful decompositions, with few components and improved interpretability compared to fixed-rank NMF and deep learning baselines.*

1. Introduction

Model pruning techniques effectively reduce the complexity of overparameterized deep networks, yet they often operate with fixed compression ratios without revealing why certain parameters are removed or offering principled stopping criteria [3, 31]. Simultaneously, interpretability remains a critical challenge, particularly in domains where understanding the model's internal representations is as important as its predictive accuracy. This is especially true for Blind Source Separation (BSS) problems, where the goal is to decompose a mixed signal into its underlying constituent sources without prior knowledge of the sources or the mixing process [44]. Extending the concept of blind separation to image data, various methods for Blind Image Decomposition (BID) were developed for applications in remote sensing, to distinguish the land cover types present [21], in medical imaging, to isolate tissue types and anatomical structures in CT scans [30], or in natural images, separating superimposed components such as rain streaks, raindrops, snow, and haze [13]. A canonical example is Nonnegative Matrix Factorization (NMF), a powerful tool for discovering parts-based representations in data, such as images [24, 25, 29]. However, standard NMF requires the factorization rank (*i.e.*, the number of components) to be specified beforehand, a critical limitation in multispectral (MS) historical document analysis, where the true number of sources is unknown (*e.g.*, the number of distinct inks, substrates, auxiliary markings, and degradation products present).



(a) MS Cube (b) Pseudo color image (c) Segment Anything [23] (d) Dinov2 + Mask2Former [33] (e) MS-Former [20] **(f) Ours**

Figure 1. Qualitative comparison of unsupervised segmentation methods with adaptive rank on sample BT56 from MSBin dataset.

Analyzing ancient documents captured under multiple wavelengths (Multispectral or Hyperspectral (HS)), is essential for revealing hidden text, distinguishing chemically similar inks, and separating overlapping content in degraded manuscripts [10, 40]. Yet, decomposing these MS images into meaningful components (*e.g.*, text, background, stains,

Email correspondence: kilian.declercq.1@ens.etsmtl.ca

stamps) remains extremely challenging due to ink diffusion, paper degradation, spectral overlap, and low signal-to-noise ratios [14], making automatic rank determination crucial for document analysis and preservation.

Even sophisticated modern segmentation approaches, including Vision Transformers (ViT), often struggle with document images. This is primarily due to the inherent fine-grained nature of text, where thin strokes and delicate structures are difficult for patch-based processing. Even SoTA segmentation transformers, presented as global and multi-task, exhibit limited performance: Dinov2 [33] may help identify document regions but struggles with character-level segmentation, while Segment Anything Model (SAM) [23] cannot find a balance between over-segmentation (splitting text into individual characters) and under-segmentation (merging multiple characters into coarse blobs), as seen on Fig. 1. These models also don't benefit from the advantages of MS imagery with a fixed number of channel design, tailored for RGB input. An ideal unsupervised method for multiband image decomposition should not only separate the layers accurately but also be interpretable (*i.e.*, revealing a clear spatial distribution for each component with its associated spectral signature) and adaptive to diverse configurations (*e.g.*, variable depth, resolution or wavelengths).

To address these limitations, we introduce **PRISM**, **P**runing for **R**ank-adaptive **I**nterpretable **S**egmentation **M**odel. This framework re-envisions pruning not only as a compression tool, but as a mechanism for dynamic rank selection within an NMF-inspired decomposition model. PRISM employs a lightweight, asymmetric convolutional autoencoder architecture. The encoder uses spatial attention to learn structured spatial abundance maps from the multiband image input. The decoder enforces interpretability through a nonnegative linear layer, reconstructing the image from these abundances and learned spectral signatures. Crucially, PRISM starts with an overcomplete set of potential components and integrates a learnable pruning mechanism that progressively deactivates weights tied to the most redundant maps during training. This process is guided by orthogonality constraints encouraging distinct components and by a criterion inspired by the Minimum Description Length (MDL) principle [11, 43]. The latter balances reconstruction fidelity with model compression (linked to still active map count) to identify the optimal rank. The convolutional structure facilitates information redistribution from pruned components to active ones, effectively merging semantically related features. PRISM thus learns both the decomposition and the optimal number of components directly from the data, yielding interpretable low-rank representations, tailored for challenging document analysis tasks.

The main contributions of this paper are as follows:

i) We introduce a new pruning approach guided by a similarity criterion with an overcomplete-to-optimal training strategy, enhancing model interpretability and segmentation quality.
ii) To preserve the spatial relationship between pixels, we implement a new lightweight hybrid NMF–convolutional autoencoder structure capable of processing whole MS and HS cubes of various resolutions and depths.
iii) Unlike existing models that integrate a two-factor NMF, we propose a novel architecture with a tri-factorization scheme and a non-local attention module for a more flexible representation and better focus on important regions.
iv) We validate PRISM on real datasets, showing its effectiveness for text binarization via MS cube decomposition. Quantitative and qualitative results confirm SoTA performance for unsupervised historical document segmentation. Generalization to remote sensing related task, demonstrates method adaptability, key for handling the unpredictable structure of historical documents.

2. Related work

PRISM integrates concepts from several research areas. We position our work relative to recent advances in (i) blind decomposition and spectral unmixing, (ii) models combining deep learning with matrix factorization principles and (iii) interpretable pruning and unsupervised rank selection.

Blind decomposition and spectral unmixing. Blind Image Decomposition (BID) seeks to separate a composite image into its constituent layers when both the sources and the mixing process are unknown [13]. This general task encompasses various applications, including removing environmental effects on images [44], decomposing images and videos into dynamic foreground layers and static backgrounds [1], or separating illumination effects for automatic white balancing [46], with recent deep learning approaches significantly advancing the area. In remote sensing and medical imaging, an analogous problem is spectral unmixing, which decomposes hyperspectral (HS) or multispectral (MS) images into a set of pure spectral signatures (*i.e.*, endmembers) and their corresponding spatial abundance maps [22, 32]. In HSI analysis, transformer-based architectures combined with diffusion models, like DiffUnmix [48], reformulate denoising and unmixing as conditional abundance generation, leveraging multi-scale attention for improved endmember extraction. Other works propose novel paradigms like "unmixing before fusion" to generate synthetic HSI data by learning generative models in a lower-dimensional abundance space [47]. While powerful, these methods often rely on large pre-trained models, require task-specific heuristics, or assume a fixed number of components. Furthermore, their application to the unique challenges of historical document images (*e.g.*, fine textual details, bleeding or overlapping inks, few annotated data, varying degradations, etc.) is not straightforward.

Hybrid Autoencoder–Factorization models. A growing body of research explores the integration of classical matrix factorization techniques, particularly NMF, with deep neural networks to leverage the strengths of both approaches. These hybrid models often aim to enhance interpretability by constraining the network's latent space or weights according to factorization principles. In the HS unmixing field, Autoencoder (AE) networks were first introduced using a pixel-wise approach (*i.e.*, input vector representing a single data point) [27]. While this early work effectively estimated abundance-like representations decoded via fully connected layers, the explicit connection to matrix factorization was not fully drawn. Subsequent research in remote sensing built upon this, introducing deeper architectures like variational AEs [45], recurrent structures such as LSTMs [50], 3D CNN for spatial-spectral feature extraction [1], and more recently incorporating ViT with spatial-spectral attention [39]. Concurrently, non-neural factorization methods like NMF and Archetypal Analysis continue to be effectively applied to HS unmixing [49, 51]. A key development in bridging unmixing AEs with convex NMF was a series of work by Palsson *et al.*, summarized in [38], interpreting the AE's decoder weights as endmembers and the latent features as abundances. To address the loss of spatial context inherent in pixel-wise approaches, a lightweight convolutional AE structure was also introduced, demonstrating strong performance, particularly in generating spatially coherent abundance maps [37]. In parallel, for MS document image decomposition, advanced related factorization concepts have proven their effectiveness, such as NMF variants with optimization on orthogonal manifolds (MA-ONMF) [41] or variational Bayesian NMF [42], which demonstrated interest in the task of model selection. AE structures have also been explored for the text binarization task [4], but no hybrid structures have been observed.

Interpretable pruning and unsupervised rank selection. Recent efforts in interpretable pruning leverage similarity metrics to identify and remove redundant network components like weights, filters, concepts, channels, or even layers [2, 5, 28]. While this offers a degree of interpretability by targeting similarity-based redundancy, such approaches, to our knowledge, have not been directly formulated to perform dynamic rank selection for unsupervised decomposition tasks. While rank selection (*i.e.*, determining optimal number of components in images) remains an open challenge in computer vision, only very few recent approaches have emerged, even for Vision Transformers (ViT), which have become foundational for segmentation tasks. Despite recent advances incorporating hierarchical clustering, methods such as Mask2Former [6] are designed for dataset-specific applications, requiring extensive pre-training on labeled data. DINOv2 [33] proposes a self-supervised model, extracting robust visual representations directly from data, allowing SoTA performances on downstream tasks even without fine-tuning. However, it relies on adding a trained segmentation head, either a linear layer or a Mask2Former head, that learns the mapping between the extracted features and the predefined labeled materials. Segment Anything Model (SAM) [23] employs a promptable ViT-based image encoder for universal image segmentation. It also proposes an image automatic segmentation with a predefined visual prompt and a masks merging strategy based on quality metrics and non-maximum suppression. However, as discussed earlier, the quality of the produced segmentation highly depends on the input prompt, requiring user involvement for alignment, and is not suitable for document images. In medical imaging, where the scarcity of large-scale annotated datasets is also a challenge, MS-Former [20] proposes a transformer architecture with self-supervisory regularization for unsupervised segmentation of skin and lung images. To the best of our knowledge, this is the only ViT-based method capable of dynamic rank selection with complete unsupervised training for multiband image segmentation.

Relation to PRISM and key novelties. PRISM builds upon these hybrid methodologies, specifically targeting document image segmentation. It employs a convolutional AE architecture, adapted from [37], combined with orthogonal NMF principles [41]. Crucially, PRISM distinguishes itself from the aforementioned fixed-rank methods by introducing a dynamic pruning mechanism to determine the optimal number of underlying factors (r). Unlike MDL-based approaches requiring post-hoc rank selection after multiple NMF runs [43], or variational Bayesian NMF methods that impose an automatic relevance determination (ARD) prior on the optimization but require per-input hyperparameter tuning [42], PRISM enables MDL-guided rank selection within a single end-to-end training process. While MS-Former [20] offers a related online adaptation for medical imaging, PRISM differs by maintaining both NMF strengths and CNN high-resolution efficiency. By integrating these elements, PRISM aims to achieve a more interpretable and adaptive decomposition of multiband images.

3. The proposed model

3.1. Problem formulation

In spectral data imaging, the Linear Mixing Model (LMM) is a widely used framework to describe the generation of MS/HS images. Under the LMM, each observed pixel in the image is assumed to arise from a convex combination of a finite set of spectral signatures (known as endmembers), perturbed by additive noise. Let $\mathbf{Y} \in \mathbb{R}_{+}^{b \times n}$ be the observed spectral data matrix, where b is the number of spectral bands and n the number of pixels. The LMM expresses the data as :

$$\mathbf{Y} = \mathbf{E}\,\mathbf{A} + \mathbf{N}, \tag{1}$$

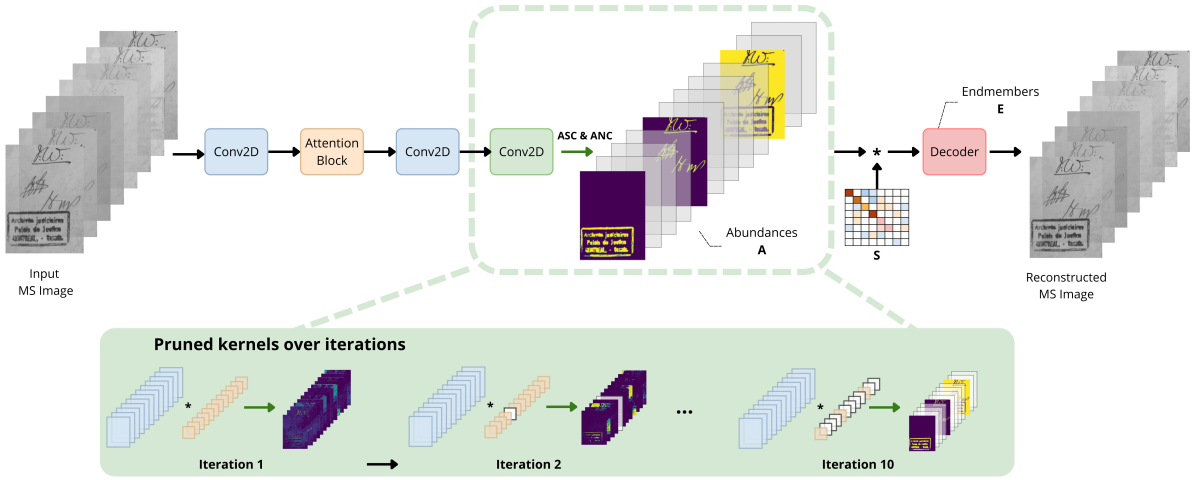


Figure 2. Overview of the proposed architecture and the iterative pruning strategy.

where $\mathbf{E} \in \mathbb{R}_+^{b\times r}$ contains r pure endmember spectra as columns, $\mathbf{A} \in \mathbb{R}_+^{r\times n}$ holds the corresponding abundance fractions, and $\mathbf{N}$ represents additive noise. To ensure that each pixel is modeled as a convex combination of endmembers, the abundance matrix $\mathbf{A}$ is subject to two constraints: the Abundance Nonnegativity Constraint (ANC), $\mathbf{A} \geq 0$, and the Abundance Sum-to-One Constraint (ASC), $\mathbf{1}_r^\top \mathbf{A} = \mathbf{1}_n^\top$, where $\mathbf{1}_k$ is the all-ones vector in $\mathbb{R}^k$.[1]

NMF can be seen as a process that seeks to reverse the linear mixing in Eq. (1) by finding nonnegative matrices $\mathbf{E} \in \mathbb{R}_+^{b\times r}$ and $\mathbf{A} \in \mathbb{R}_+^{r\times n}$ that satisfy

$$\mathbf{Y} \approx \mathbf{E}\,\mathbf{A}, \tag{2}$$

often under additional constraints—such as sparsity on $\mathbf{A}$, smoothness on $\mathbf{E}$, or orthogonality of columns—to enhance endmember and abundance estimation following prior knowledge (*e.g.*, abundances are sparse or endmember spectra vary smoothly across bands). To fit complex data while retaining the same constraint-based philosophy, an additional matrix $\mathbf{S} \in \mathbb{R}_+^{r\times r}$ is incorporated in Eq. (2), which results in a tri-factorization model that writes

$$\mathbf{Y} \approx \mathbf{E}\,\mathbf{S}\mathbf{A}, \tag{3}$$

This tri-factorization thus extends constrained NMF, offering extra flexibility to model nonlinear or higher-order interactions while preserving the interpretability guaranteed by nonnegativity [8]. In contrast to most existing methods [1, 37, 45, 50], we adopt model (3) in our architecture as the intermediate scaling matrix $\mathbf{S}$ relaxes the optimization, providing additional flexibility by encoding the latent relationships among endmembers ($\mathbf{E}$) and the abundance ($\mathbf{A}$).

[1]Treating $\mathbf{Y}$ as nonnegative is natural since pixel intensities, whether radiance, reflectance, or even photon counts, are inherently nonnegative.

3.2. Proposed PRISM model

The proposed autoencoder-like NMF model, **PRISM**, consists of an encoder-decoder structure with a non-local attention layer and an interpretable pruning mechanism designed to automatically estimate the number of components found in the input MS scene during training. The encoder is designed to extract a rich feature representation from an input MS cube, $\mathbf{Y} \in \mathbb{R}^{h\times w\times b}$, where $h \times w$ is the spatial size and b is the number of spectral bands. It generates the abundance maps $\mathbf{A} \in \mathbb{R}^{h\times w\times r}$, where each map represents the relative proportions of spectral endmembers present at each pixel. Here, the rank r is initialized to a large number, with maps being progressively deactivated by the pruning process. Following Eq. (3), the decoder combines the activated abundance with a learned scale matrix $\mathbf{S}$. It then reconstructs the input image with learned endmembers $\mathbf{E}$. An overview of the whole architecture is illustrated in Fig. 2.

Encoder. The encoder consists of four sequential layers designed to process the entire set of MS bands simultaneously. In contrast to existing approaches [37] that adopt a patch-based strategy, our encoder operates on the full image, extracting useful relationships from all pixels. It begins with a 2D convolutional layer employing $c = 4 \times b$ kernels. The resulting feature maps are fed into an attention block, which refines the feature representation by emphasizing informative regions. The encoder concludes with another depthwise 2D layer, followed by a pointwise convolution, creating an expand-and-compress pattern that captures spatial and channel-wise dependencies before outputting r maps. Finally, a temperature-scaled softmax function is used to handle the nonnegativity of the abundances, elegantly satisfying both ANC and ASC conditions. By adjusting the temperature hyperparameter, the sparsity of abundance coefficients can be tailored to match the unmixing requirements (*i.e.*, mixed materials for remote sensing unmixing or pure pixels for text segmentation).

Decoder. In contrast to models utilizing symmetric encoder-decoder structures like [1], our PRISM architecture incorporates a lightweight, nonnegative decoder that preserves the interpretability of its learned components. This decoder reconstructs the b input spectral bands from the r abundance maps ($\mathbf{A}$) by emulating the tri-factorization $\mathbf{Y} \approx \mathbf{ESA}$. Specifically, the r abundance maps are first processed by an interaction matrix $\mathbf{S}$, which is a learnable $r \times r$ nonnegative matrix. The resulting r transformed maps then pass through a final unbiased pointwise convolution, whose $b \times r$ nonnegative weights constitute the learned endmember matrix $\mathbf{E}$ and produce the b output bands. This streamlined, asymmetric design significantly reduces computational demands and the total parameter count; for instance, the decoder's learnable parameters for both $\mathbf{S}$ and $\mathbf{E}$ totalled $r(r + b) = 2088$ for a 162-band hyperspectral image, before pruning. Both $\mathbf{E}$ and $\mathbf{S}$ are constrained to be nonnegative and are updated via backpropagation.

Attention block. The core of our attention block is adapted from the original Large Kernel Attention (LKA) [12] that is designed to enhance feature representation by capturing both local and global relationships between pixels of MS images. The module begins by processing the input feature map through a standard 2D convolutional layer (DW-Conv) with a 5×5 kernel size to extract local spatial features. This is followed by a dilated depth-wise convolution (DW-D-Conv) with a 7×7 kernel, which expands the receptive field without increasing parameters, capturing broader context by processing each channel independently. A pointwise convolution then combines features across all channels, refining the representation. The resulting feature map is multiplied element-wise with the original input feature map, acting as an attention mechanism that emphasizes informative regions and suppresses less relevant features. Finally, a pointwise convolution, with the same number of output channels as the input, further refines the features, producing the final output. Attention block is detailed in Supp. Mat.

Interpretable pruning. At the initial state, the model is deliberately overcomplete, with more abundance maps than objects, enabling the network to capture a comprehensive set of all potential components, as done in [20, 42]. From these starting r_{init} abundance maps, an iterative training strategy is employed where the model alternates between training until early stopping, and a pruning step that deletes the least informative map. At each pruning step, the model is forced to adapt itself through merging, shifting, or removal of elements; effectively distilling its knowledge into a more compact representation. This strategy offers the significant advantage of providing interpretability into the model's focus and enabling control over its decision-making process. To identify the least informative map, a pairwise similarity score between all unpruned maps is calculated to identify the most similar pair. From this pair, the map with the smallest Frobenius norm (*i.e.*, the map representing the smallest image proportion) is added to the set of pruned maps. Each map in this set sees the pointwise convolution weights producing it being masked, as for its corresponding decoder weights, which represent the associated endmember. The similarity formula we developed is adapted from the cross-correlation coefficient. To mitigate an object fragmentation side effect, which may be induced in early overcomplete steps, a spatial smoothing operation is applied prior to calculation, using a Gaussian-like filter. We observed that if two maps represented similar spatial regions, having a small object within a larger one was often more penalized than the same object split into two separated equal parts. Therefore, the difference between the two maps activation was included in the equation. Moreover, the spectral angular distance between the corresponding endmembers was incorporated to penalize materials having similar endmembers but located at different spatial positions. The final similarity formula, capturing both spatial and spectral similarity between materials, is then expressed as follows:

$$\mathcal{S}_{i,j} = \frac{1}{|\langle \mathcal{A}_i - \mathcal{A}_j, \mathbf{1} \rangle| + \epsilon} \frac{\langle \mathcal{A}_i, \mathcal{A}_j \rangle}{\|A_i\| \|A_j\|} + \lambda_{\text{SAD}'} \mathcal{L}_{\text{SAD}'}(E_i, E_j), \quad (4)$$

where $\mathcal{L}_{\text{SAD}'}(E_i, E_j)$ is the spectral angular similarity between endmembers, $\lambda_{\text{SAD}'}$ is a parameter added to scale this distance, $\langle \cdot, \cdot \rangle$ is the inner product, and ϵ prevents division by zero (see Supp. Mat. for ablation and hyperparameters).

Objective function. The loss function is composed of three terms. The reconstruction error between the input MS cube and the generated output is measured using a combination of Spectral Angle Distance (SAD) and MSE. SAD, a scale-invariant measure, addresses spectral variability by evaluating differences along the depth dimension [38], while MSE enforces stricter material discrimination by encouraging element-wise reconstruction. A third term, $\mathcal{L}_{\text{orth}}$, enforces the orthogonality of the abundance matrix $\mathbf{A}$, to promote spatially distinct and localized segments. The total loss formula, detailed in Supp. Mat., is given as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{SAD}} + \lambda_1 \mathcal{L}_{\text{MSE}} + \lambda_2 \mathcal{L}_{\text{orth}}. \quad (5)$$

Rank selection criterion. Our channel pruning approach seeks to minimize an objective function grounded in the Minimum Description Length (MDL) principle [11, 43]

$$\min_H \{L(H) + L(D \mid H)\}, \quad (6)$$

Here, $L(D \mid \mathcal{H})$ quantifies the data reconstruction error, given a model $\mathcal{H}$, while $L(\mathcal{H})$ represents the model's complexity or description cost. This framework naturally relates to optimizing our Tri-NMF-like model ($Y \approx ASE$). In this analogy, the model $\mathcal{H}$ encompasses the factors $\{A, S, E\}$.

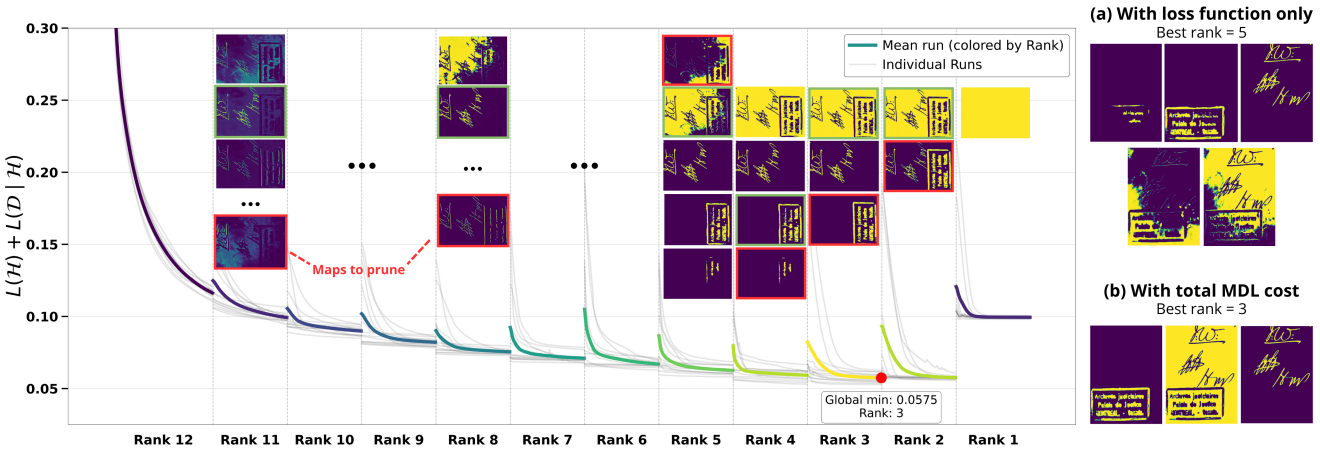


Figure 3. Evolution of $L(H) + L(D \mid \mathcal{H})$ during pruning iteration on a MStex sample with 12 initial maps. Reference abundance maps are displayed for some ranks. The loss value decreases with each iteration until the model reaches an optimal decomposition for 3 materials.

The reconstruction term, $\mathcal{L}_{\text{SAD}} + \lambda_1 \mathcal{L}_{\text{MSE}}$, directly corresponds to $L(D \mid \mathcal{H})$. The model complexity $L(\mathcal{H})$ is influenced by both the structure and size of the factors, the latter explicitly minimized during the iterative pruning with dimensionality reduction (*e.g.*, for A and E). The structural complexity is captured through the orthogonality constraint $A^T A = I$, which favors more compact representations. We further penalize model size with an additional cost proportional to rank r, scaled by a factor λ_3. This complete objective function therefore represents the total MDL cost, $L(\mathcal{H}) + L(D \mid \mathcal{H})$, summing the data reconstruction error with the model complexity (comprising both the structural cost influenced by orthogonality, $\lambda_2 \mathcal{L}_{\text{orth}}$ and the rank-dependent size penalty $\lambda_3 r$). Figure 3 plots this total MDL cost across successive pruning stages on a sample image. It reveals a characteristic curve that descends, as pruning reduces the rank while the model adapts its features to better reconstruct the data D, before reaching a global minimum. Beyond this point, the function starts to increase, indicating the onset of overfitting, with the model forced to merge components identified as most similar. This empirical minimum represents an optimal MDL trade-off; it balances compression with data fidelity, identifying the most relevant component set, and serves as our rank selection criterion. Ablation comparing reconstruction loss alone versus total MDL cost is shown in the Supplementary Material.

4. Experiments

We conducted our main experimental evaluation on 51 distinct MS input cubes from MSTEx-1, MSTEx-2, and MS-Bin collections, as well as from the Urban HS dataset.

MSTEx 1 & 2 [14] These collections include 20 MS cubes of historical manuscripts from the 17th to 20th centuries, provided by BAnQ (Bibliothèque et Archives nationales du Quebec). Each cube consists of 8 spectral bands captured in the range from 340nm to 1100nm. These images were acquired using a Chroma X3 KAF 6303E (Kodak) camera at a 6-megapixel resolution (9 µm pixel size). Ground-truth (GT) classifies pixels into text or non-text categories only.

MSBin [17] This collection contains 30 "test" MS cubes captured using a Phase One IQ260 achromatic camera with a resolution of 60 megapixels, across twelve narrow-band ranging from 365nm to 940nm. The images were digitized from two manuscripts: the Bitola-Triodion ABAN 38 (Book BT) and Enina-Apostolus NBMK 1144 (Book EA), the latter in severely degraded condition. The ground truth identifies pixels as text foreground (iron gall ink), background, or uncertain regions, too complex for manual annotations (the latter not included in evaluation). Some images also feature a second red ink foreground, as seen on Fig. 1.

Urban [19] This dataset is a cornerstone in hyperspectral unmixing research, which features 307×307 pixels with 162 spectral bands. Acquired by the HYDICE sensor over the Washington DC Mall, it provides three GT scenarios with 4, 5, and 6 materials, ideal for benchmarking our rank selection algorithm. With a 2×2 m² spatial resolution and a spectral range of 400–2500 nm, it supports detailed material identification in urban scenes with highly mixed sources.

4.1. Text extraction analysis

4.1.1. Compared methods

Given that GT exists only for text components, we employ text binarization benchmarks to assess decomposition quality. As a baseline, we use Howe's method [18], which is one of the SoTA document image binarization algorithms in the literature. We also compare the performance of five traditional models designed for text extraction: Spatial Kernel K-Harmonic Means (SKKHM) [26], Gaussian Mixture Models (GMMs) [16], Selectional AE (SAE) [4], and Adaptive Coherence Estimator (ACE) v1 & v2 [7, 15]. We also compare against two NMF models for BID, namely, MA-ONMF, a bi-orthogonal NMF [41], and VBONMF, a Vari-

ational Bayesian Orthogonal NMF [42]. Additionally, we consider cross-domain methods, with several LMM-based AE for unmixing: Endnet [34], MTAEU [36], DAEU [35], OSPAEU [9], and CNNAEU [37]. We also evaluated MS-Former [20], a promising adaptive-rank ViT-based model.

4.1.2. Metrics

To objectively evaluate the performance of the proposed model and quantitatively compare it with state-of-the-art methods, we employed four standard evaluation metrics: F-Measure (FM), Distance Reciprocal Distortion (DRD), Negative Rate Metric (NRM), and Peak Signal-to-Noise Ratio (PSNR). FM and PSNR values are expressed as percentages (%) and decibels (dB), respectively, with higher values indicating better binarization quality. Conversely, DRD and NRM values range from 0 to 1, with lower values indicating superior binarization performance. Detailed metric formulas are provided in Supplementary Material.

4.1.3. Text extraction results

Experimental results on the 20 cubes from MSTEx-1 and MSTEx-2, seen in Tab. 1, confirm our model's efficacy, achieving the best performance across most metrics. This underscores its strong image decomposition capabilities. While MA-ONMF, VBONMF, and ACE2 also demonstrated competitive results, their average performance was lower than ours. Traditional methods such as SKKHM and SAE yielded limited performance, in some cases even underperforming Howe binarization. MS-Former recorded the lowest overall performance on these datasets. Notably, some HS unmixing autoencoder-based methods, specifically OSPAEU and MTAEU, showed results comparable to document-specific techniques without task-specific tuning.

Table 1. Average performance across the 20 MS cubes in MSTEx-1 & 2.

Method	**FM**(%)↑	**DRD**(10^{-3})↓	**NRM**(10^{-2})↓	**PSNR**↑
Cross-domain methods				
MSFormer [20]	57.15	19.76	16.79	11.55
Endnet [34]	61.23	32.20	14.08	11.87
CNNAEU [37]	65.43	31.02	17.99	13.22
DAEU [35]	71.49	12.33	14.88	14.50
OSPAEU [9]	73.62	18.74	12.98	14.78
MTAEU [36]	73.30	17.01	8.30	13.95
Document-tailored methods				
SAE [4]	64.98	9.94	14.33	13.45
SKKHM [26]	71.78	11.09	13.37	14.35
Howe [18]	76.96	6.63	8.94	14.94
GMM [16]	80.72	5.12	10.42	16.05
ACE v1 [15]	83.85	4.12	9.11	-
MAONMF [41]	85.09	3.59	7.52	-
ACE v2 [7]	85.15	3.66	8.29	-
VBONMF [42]	**85.70**	**3.55**	**6.31**	**17.12**
Ours	**86.47**	**3.11**	**7.30**	**17.24**

On the 30 input cubes of MSBin (Table 2), our method again achieved the best results across all metrics for unsupervised text extraction. MA-ONMF and VBONMF methods, which require input-specific hyperparameter tuning, could not be reproduced for generalization to this dataset. Many document-specific methods, including Howe and SAE, exhibited diminished performance on MSBin compared to MSTEx, attributable to highly degraded samples from the EA manuscript. In some instances, these methods failed to extract any text, emphasizing the criticality of robust decomposition (see Supp. Mat.). Conversely, while achieving numerically lower metrics, cross-domain methods maintained consistent performance, indicating good generalization and validating LMM as a strong framework for both HS and MS image decomposition. MTAEU secured the second-best NRM score after our method. MS-Former showed improved performance, potentially due to the larger character strokes benefiting its patch-based architecture.

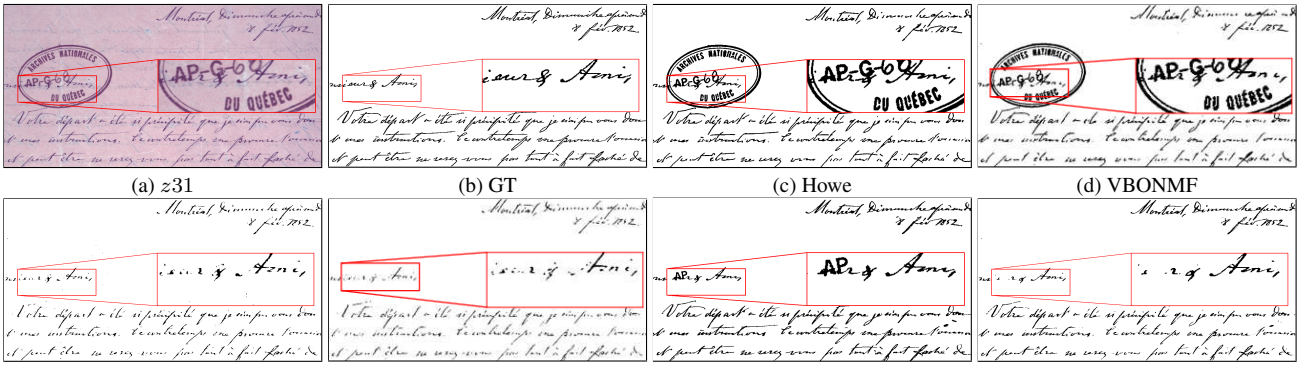
Table 2. Average performance across the 30 MS cubes in MSBin.

Method	**FM**(%)↑	**DRD**(10^{-3})↓	**NRM**(10^{-2})↓	**PSNR**↑
Cross-domain methods				
CNNAEU [37]	67.10	39.52	19.35	10.32
Endnet [34]	67.62	41.71	20.61	10.57
MSFormer [20]	69.87	46.62	12.55	13.18
OSPAEU [9]	71.59	52.90	13.14	11.77
DAEU [35]	72.68	29.73	16.50	11.77
MTAEU [36]	74.02	34.25	**12.81**	11.35
Document-tailored methods				
MAONMF [41]	-	-	-	-
VBONMF [42]	-	-	-	-
SAE [4]	40.83	50.86	32.42	8.95
Howe [18]	41.50	38.68	31.27	10.48
SKKHM [26]	73.42	26.78	17.43	11.74
GMM [16]	80.00	**20.35**	-	13.18
ACE v2 [7]	81.25	20.80	-	13.27
ACE v1 [15]	**81.28**	22.03	-	**13.28**
Ours	**82.57**	**18.71**	**10.45**	**13.74**

4.2. Qualitative analysis

4.2.1. Text component

For a qualitative assessment, Figs. 1 and 4 compares our decomposition and text extraction with different methods. This subjective analysis is essential for capturing nuances, like fine detail preservation, stroke continuity, and character clarity, that quantitative metrics may not fully reflect. In Fig. 4, most methods, including VBONMF, struggle to distinguish the overlapping stamp from handwritten text, except for OSPAEU, MA-ONMF, and our model, confirming PRISM's effectiveness. On this specific sample, GMM also demonstrates a performant text extraction, with the exception of a small stamp part, resulting in a lower FM score.



(a) $z31$ (b) GT (c) Howe (d) VBONMF

(e) OSPAEU (f) MA-ONMF (g) GMM (h) **Ours**

Figure 4. Text components extracted by different methods, ordered by FM, on the sample z31 from MSTEx-1. Visualized as in [41, 42].

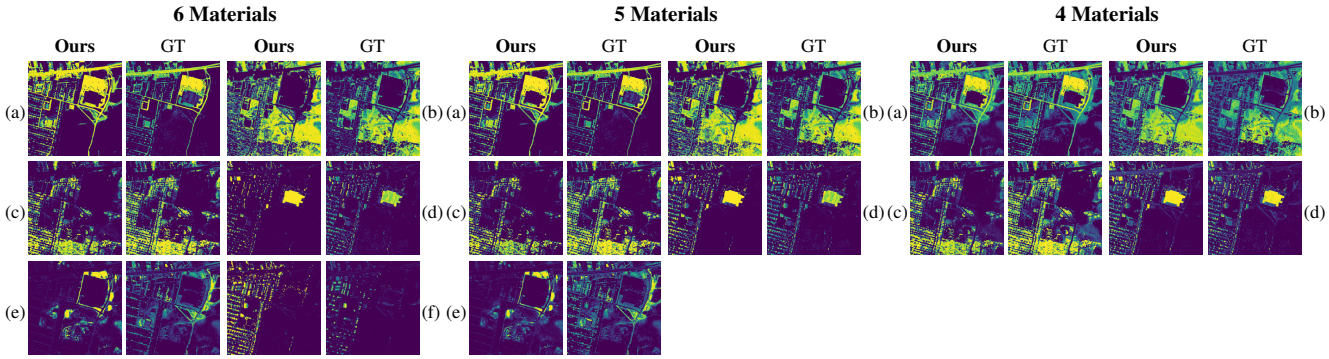


Figure 5. Qualitative assessment of PRISM on Urban dataset for 6, 5 and 4 materials: (a) Asphalt, (b) Grass, (c) Trees, (d) Roof, (e) Soil, and (f) Metal. Components (f) Metal and (e) Soil are iteratively merged to the most related maps following the three given GT scenarios.

4.2.2. Remote sensing generalization

Following the similarity with the HS unmixing task, generalization of PRISM was tested on the Urban dataset. Our model successfully extracted the abundances corresponding to three given GT scenarios for 6, 5, and 4 materials, respectively, with performance similar to fixed-rank models [38]. From twelve initial maps, $(12-r)$ abundances are generated after r pruning iterations, iteratively producing the results seen in Fig. 5. To our knowledge, this is the first approach trying to obtain the results for the three different scenarios in a single end-to-end training (more results in Supp. Mat.).

5. Discussion

This work demonstrates that similarity-driven pruning can serve as an effective rank-selection mechanism inside an NMF-constrained autoencoder.

On two challenging historical-manuscript datasets with diverse temporal and geographical origins, **PRISM** consistently outperforms classical and transformer-based methods, while producing interpretable abundance/endmember pairs. The empirical success across domains suggests that linear-mixing-based priors, when combined with lightweight deep models, remain highly competitive for data-scarce decomposition problems. The proposed approach offers significant potential for applications requiring interpretable decomposition with minimal supervision, such as cultural preservation or environmental monitoring.

Nevertheless, the MDL trade-off between compactness and semantic completeness is sensitive to the pruning schedule; aggressive compression may discard rare but meaningful components. Future versions should therefore integrate domain cues (*e.g.*, either weak labels or physics-based priors) to prevent such failures, especially for sensitive domains.

Future work will explore semi-supervised versions to incorporate domain knowledge, investigate reinforcement learning for more adaptive pruning policies, and extend the framework to natural RGB images, using PRISM as a lightweight segmentation head on top of frozen pre-trained encoders. In this context, the feature maps extracted by these encoders can be interpreted as multiband images, making our approach directly applicable for feature-space unmixing. Finally, developing robust metrics for the semantic quality of unsupervised decompositions would further solidify the performance comparison between such models.

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